

DWBI Assignment 2



Sri Lanka Institute of Information Technology

DWBI Year 3, Semester 1 - 2026

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Step 1: Data source for the Assignment 2

1. Dataset Overview

The dataset used for this project contains 50,000 hospital admission records collected over a two-year period (January 2021 – December 2022) from multiple major hospitals in Sri Lanka. Each record represents a single patient admission event and includes medical, operational, and billing information. The dataset was originally structured as a raw OLTP transactional dataset and was transformed into a data warehouse to support analytical processing and decision-making.

2. Business Scenario (your part is good)

The dataset simulates real-world hospital operations including patient admissions, department-level treatments, doctor assignments, laboratory testing, room allocation, and billing processes. This enables analysis of hospital performance, cost management, and patient trends.

3. Data Warehouse Description

The data warehouse was designed using a **star schema**, where a central fact table is connected to multiple dimension tables. The ETL process was implemented using SSIS to extract data from different sources, transform it into a consistent format, and load it into the data warehouse.

4. Fact Table

The central fact table is **Fact_Admissions**, which stores quantitative measures related to hospital admissions.

Measures

- Total_Bill_LKR
- Medication_Cost_LKR
- Room_Charges_LKR
- Lab_Charges_LKR
- Doctor_Fee_LKR
- Length_of_Stay_Days
- Lab_Tests_Count

These measures are used to analyze hospital revenue, costs, and patient stay durations.

The grain of the Fact_Admissions table is defined as one record per patient admission.

This ensures that each row represents a single admission event, allowing accurate aggregation and analysis.

5. Dimension Tables

Several dimension tables provide descriptive context for analysis:

- Dim_Date – Supports time-based analysis (Year, Month, Quarter, Date)
- Dim_Hospital – Contains hospital location details (Province, District, Hospital Name)
- Dim_Department – Represents hospital departments
- Dim_Doctor – Stores doctor details including specialty
- Dim_Patient – Contains patient demographic details
- Dim_Diagnosis – Includes diagnosis information
- Dim_Payment – Stores payment and insurance details

6. Additional Design Features

1. Hierarchies

The data warehouse supports hierarchical structures within dimensions to enable efficient OLAP operations such as drill-down and roll-up.

The following hierarchy is defined:

- Date Hierarchy (Dim_Date):

Year → Quarter → Month → Date

These hierarchy allow users to analyze data at different levels of granularity. For example, users can view total hospital revenue at the yearly level and drill down to quarterly, monthly, and daily levels. Similarly, hospital performance can be analyzed from province level down to individual hospitals.

7. Relationships (SHORT BUT POWERFUL)

The fact table is linked to all dimension tables through foreign keys, forming a star schema that enables efficient multidimensional analysis such as slicing, dicing, and drill-down operations.

8. Suitability for BI

The data warehouse supports various analytical queries such as:

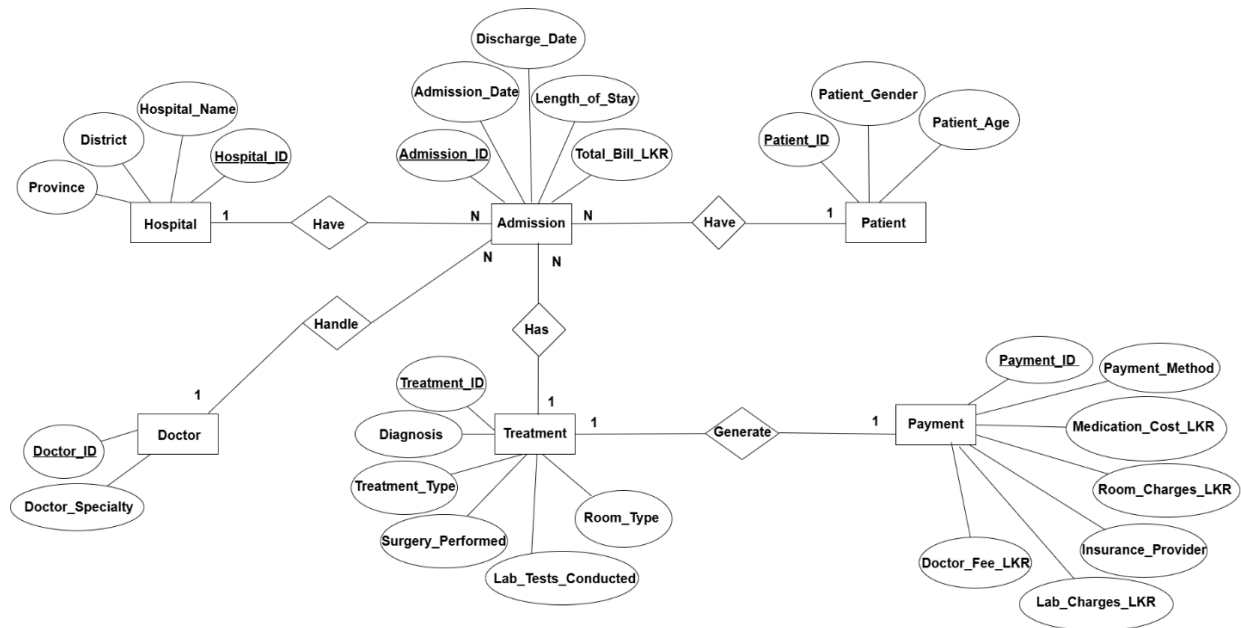
- Monthly admission trends
- Revenue analysis by hospital and department
- Cost breakdown analysis
- Length of stay analysis
- Payment method comparison
- Doctor performance evaluation

This makes it highly suitable for OLAP and Business Intelligence applications.

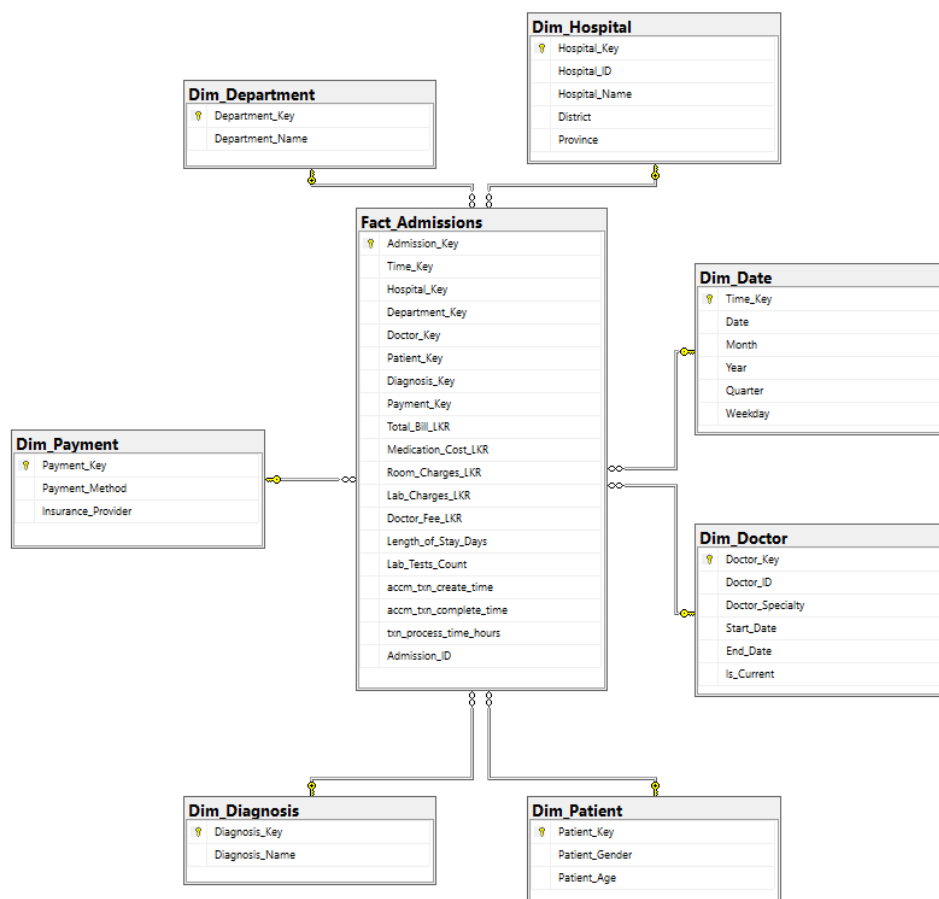
The original operational database structure is represented using an Entity-Relationship (ER) diagram. This diagram illustrates the relationships between entities such as Hospital, Admission, Patient, Doctor, Treatment, and Payment.

- A hospital can have multiple admissions.
- Each admission is associated with one patient.
- Admissions are handled by doctors.
- Each admission includes treatment details.
- Treatments generate payment records.

This ER model represents a normalized transactional system used for day-to-day operations.



ER – Diagram



Star schema

The above diagram represents the star schema design of the hospital data warehouse. The central fact table, Fact_Admissions, stores quantitative measures related to hospital admissions such as billing and length of stay.

It is connected to multiple dimension tables including Dim_Date, Dim_Hospital, Dim_Doctor, Dim_Patient, Dim_Department, Dim_Diagnosis, and Dim_Payment, which provide descriptive context for analysis.

This schema enables efficient multidimensional analysis such as revenue analysis, patient trends, and hospital performance evaluation.

Overall, the data warehouse design provides a scalable and efficient structure for multidimensional analysis and supports the development of OLAP cubes and BI reports in Assignment 2.

Step 2: SSAS Cube implementation

SQL Server Analysis Services (SSAS) was used to create a multidimensional cube based on the hospital data warehouse. The cube enables efficient analysis of large datasets using dimensions and measures.

DATA SOURCE CREATION

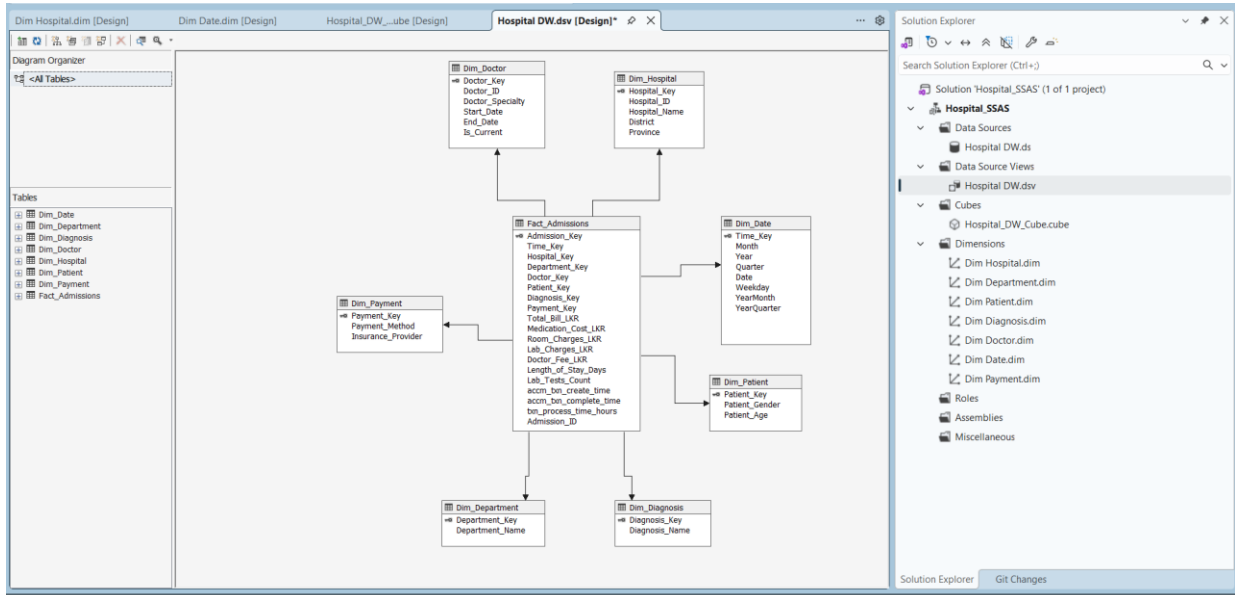
The screenshot shows the 'Data Source Designer' window with the following details:

- Data source name:** Hospital DW
- Provider:** SQLOLEDB.1
- Connection string:** Provider=SQLOLEDB.1;Data Source=ABCD\MSSQLSERVER
- Data source references:**
 - ☐ Maintain a reference to another object in the solution
 - Dropdown: Create a data source based on an existing data source
- Isolation:** ReadCommitted
- Query timeout (in seconds):** 0
- Maximum number of connections:** 10
- Data source description:** (Empty text box)

Buttons at the bottom: OK, Cancel, Help.

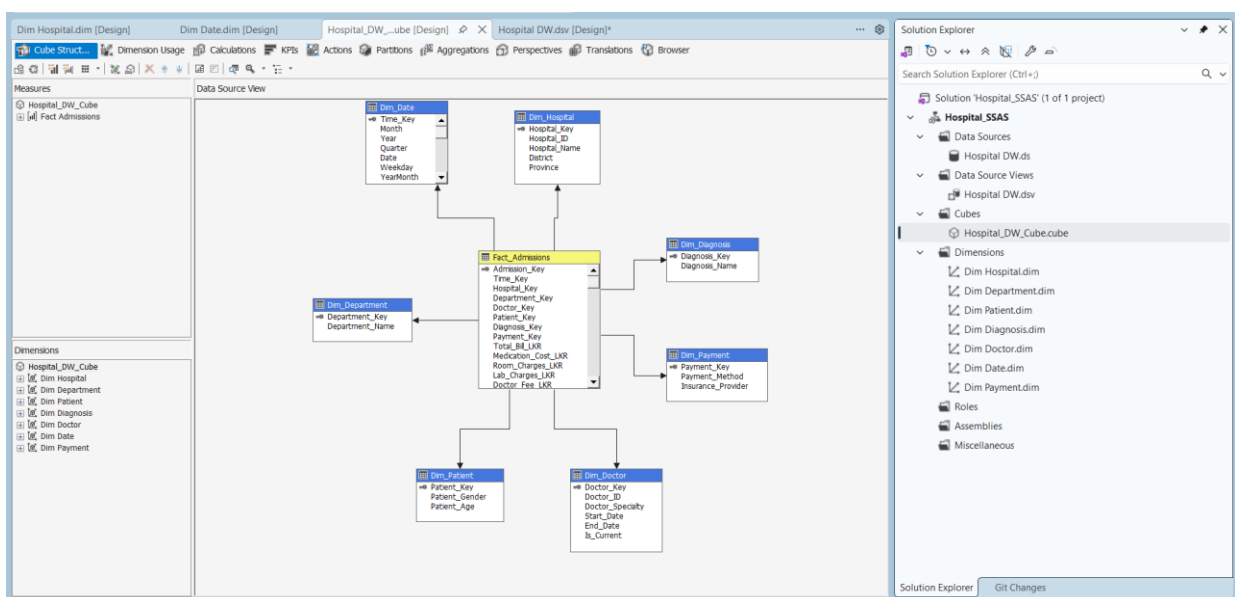
A data source was created in Visual Studio by connecting to the SQL Server instance (localhost\MSSQLSERVER2) and selecting the HospitalDW database. This connection allows SSAS to access data from the data warehouse.

DATA SOURCE VIEW (DSV)

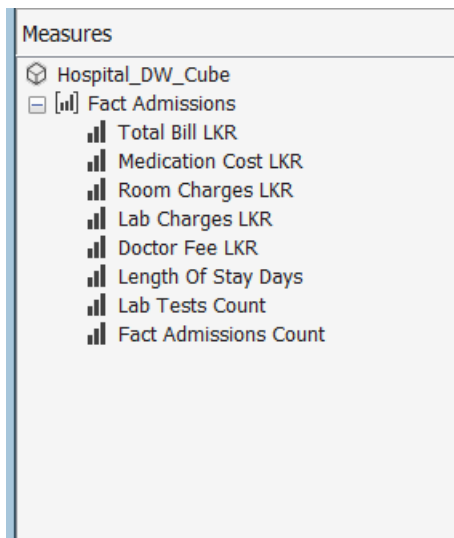


A Data Source View (DSV) was created by selecting relevant tables including Fact_Admissions and dimension tables such as Dim_Date, Dim_Hospital, Dim_Doctor, Dim_Department, Dim_Patient, Dim_Diagnosis, and Dim_Payment. Relationships between tables were automatically detected.

CUBE CREATION

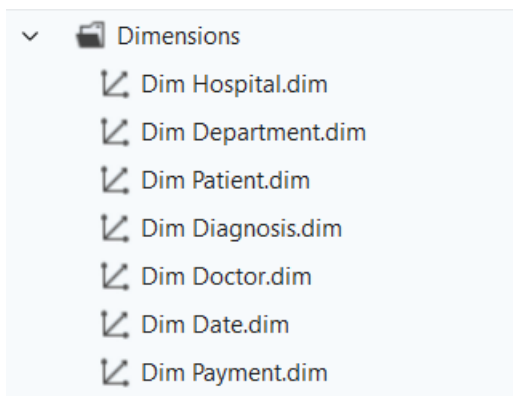


MEASURES



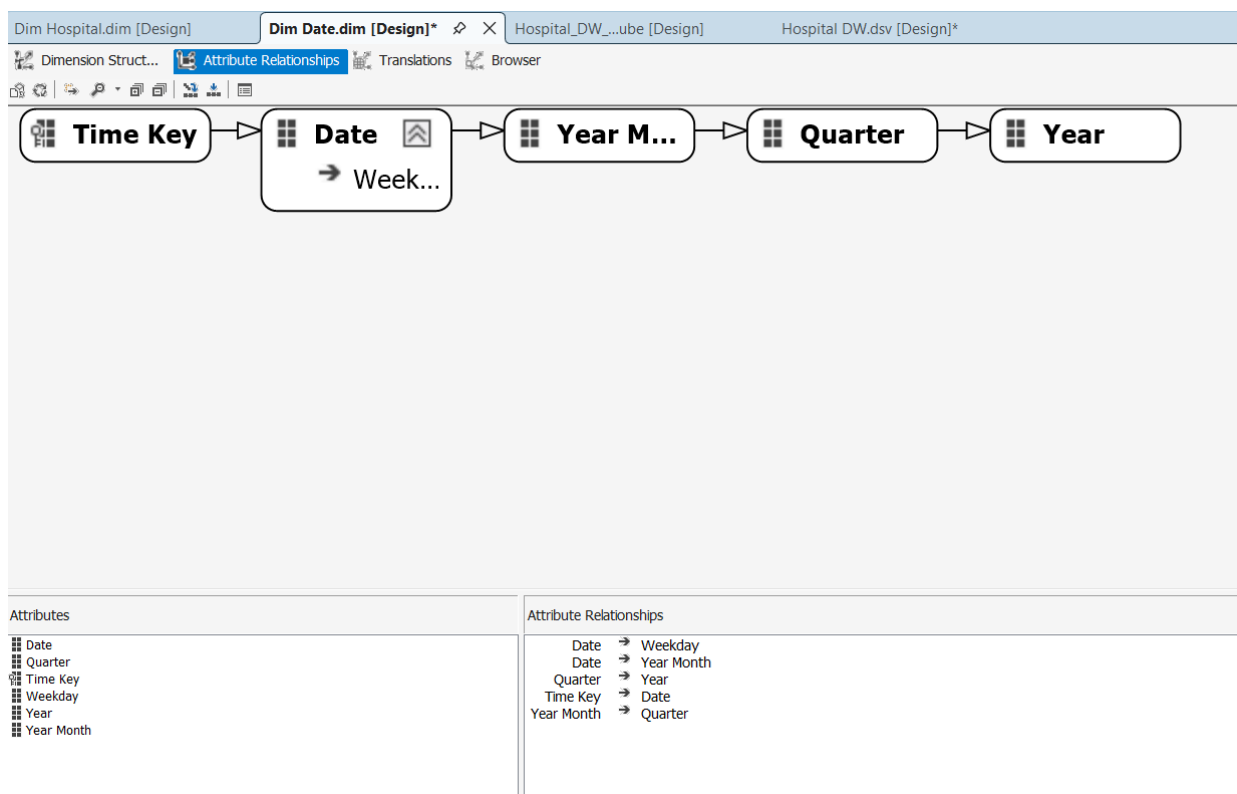
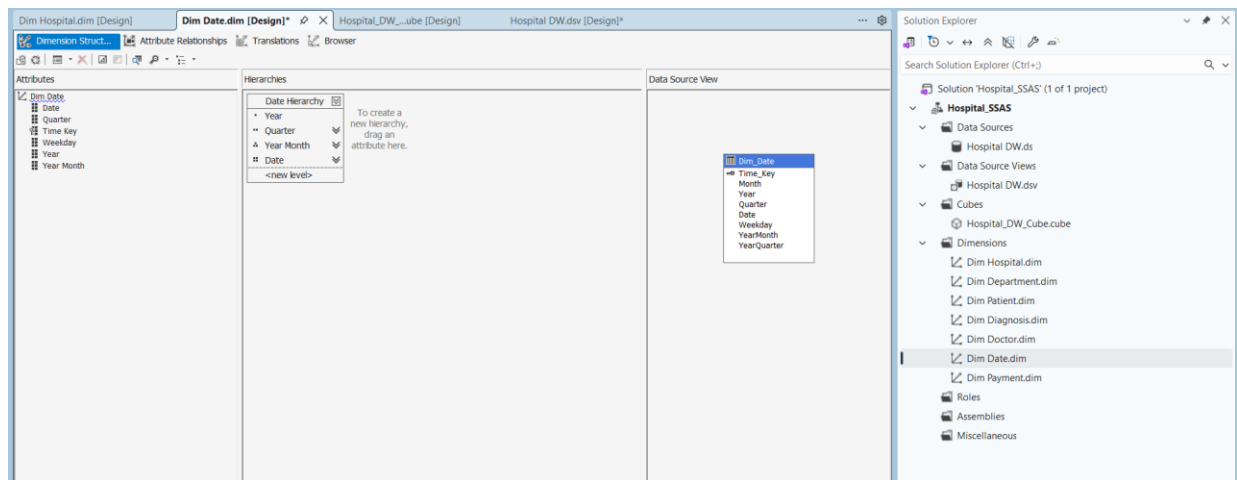
Measures were selected from the Fact_Admissions table to support analysis. These include Total_Bill_LKR, Medication_Cost_LKR, Room_Charges_LKR, Lab_Charges_LKR, Doctor_Fee_LKR, Length_of_Stay_Days, and Lab_Tests_Count. These measures allow analysis of revenue, costs, and operational performance.

DIMENSIONS



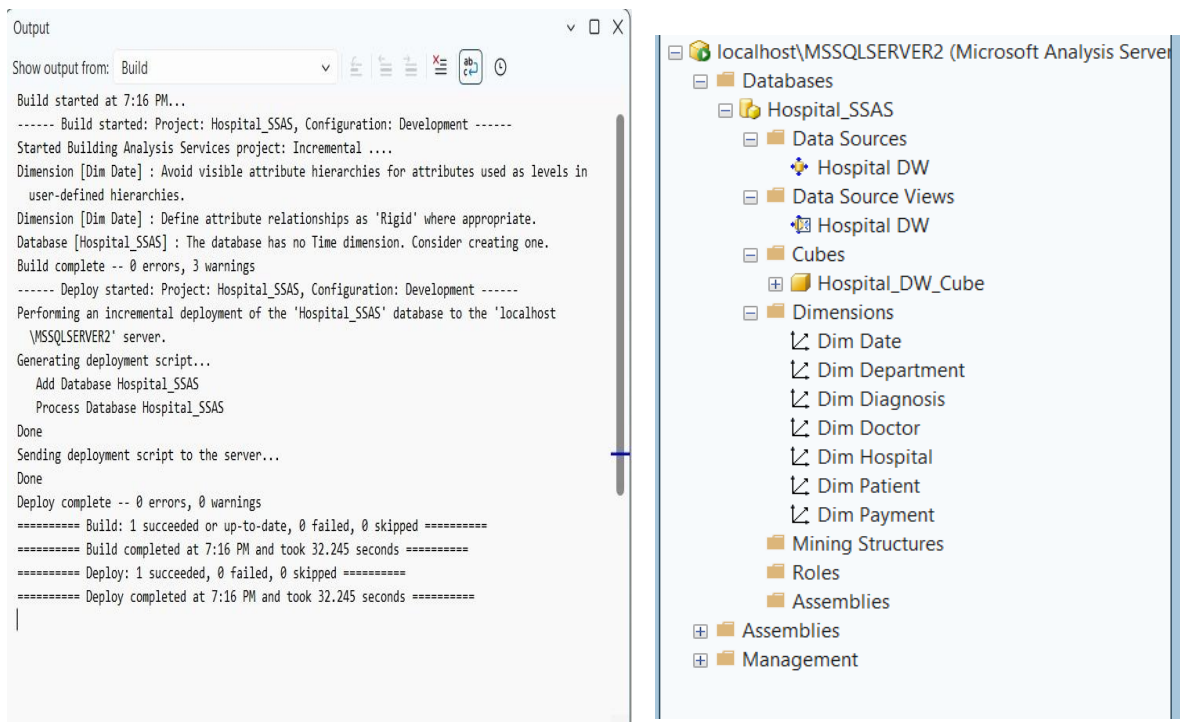
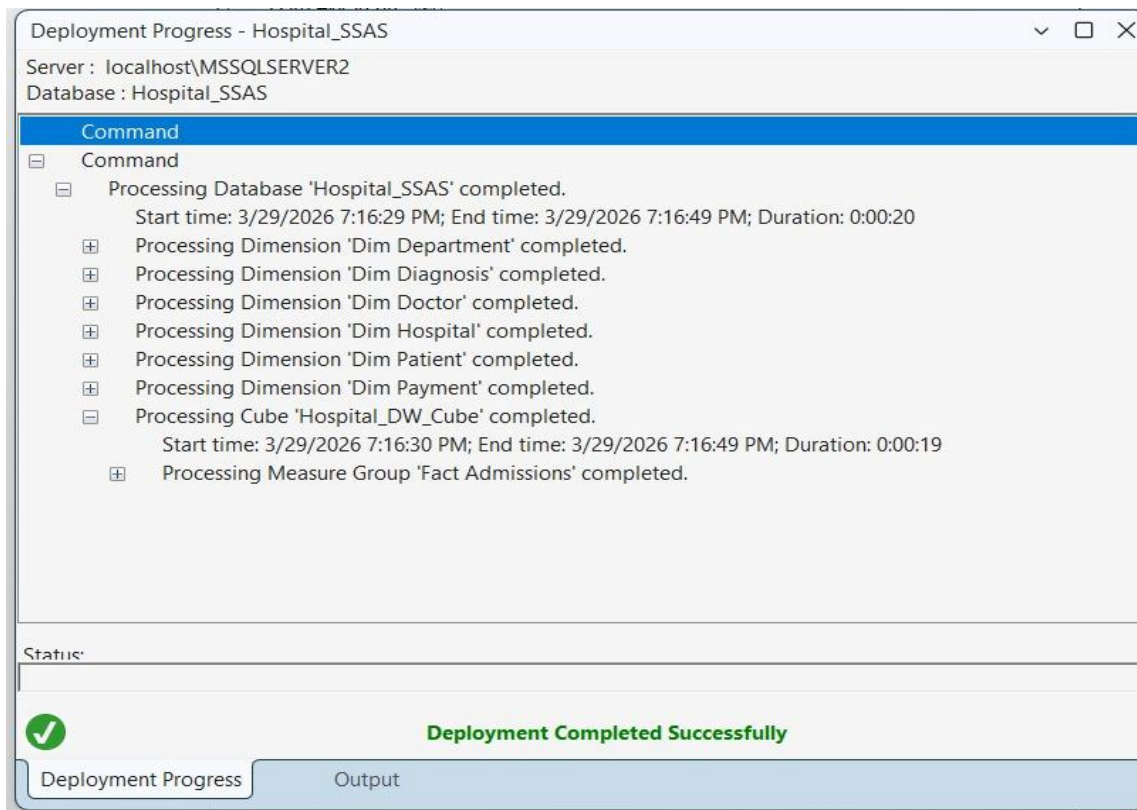
Dimensions were created from the corresponding dimension tables to provide descriptive context for analysis. These include Dim_Date, Dim_Hospital, Dim_Doctor, Dim_Department, Dim_Patient, Dim_Diagnosis, and Dim_Payment.

HIERARCHY



A hierarchy was created in the Dim_Date dimension to support drill-down analysis. The hierarchy includes Year, Quarter, Month, and Date levels. This enables users to analyze data at different levels of granularity.

CUBE DEPLOYMENT



The cube was deployed to the SQL Server Analysis Services instance (localhost\MSSQLSERVER2). Deployment processes the cube and makes it available for querying and analysis.

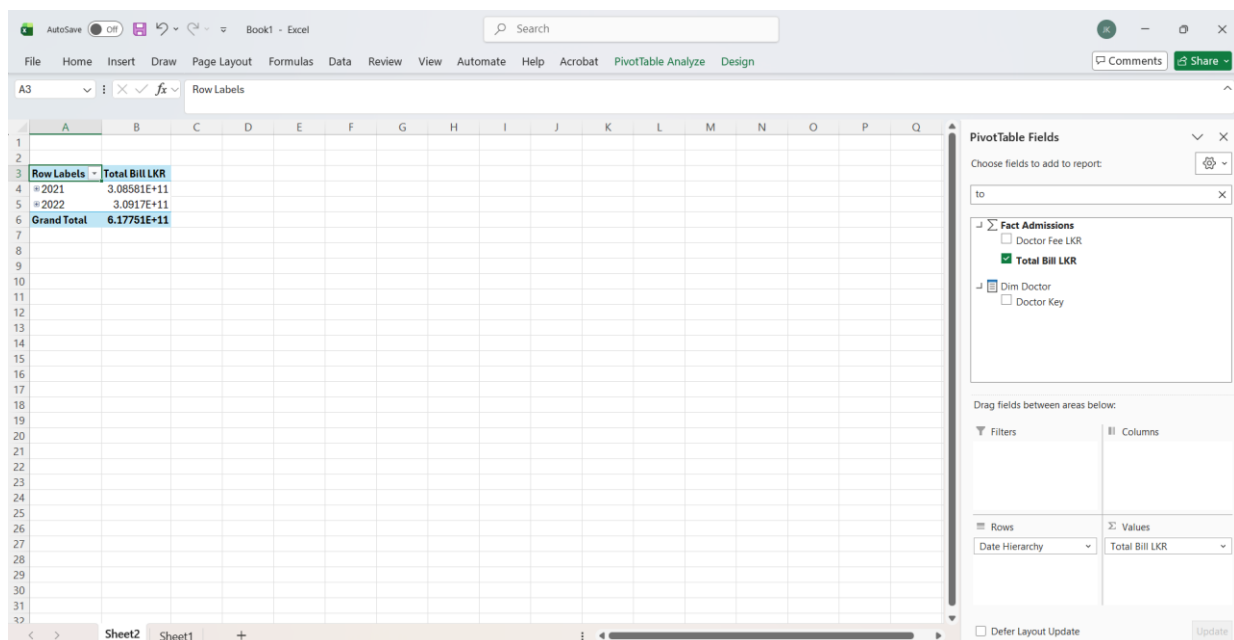
The SSAS cube was successfully created and configured using measures, dimensions, and hierarchies. The cube supports efficient multidimensional analysis and enables users to perform advanced analytical operations.

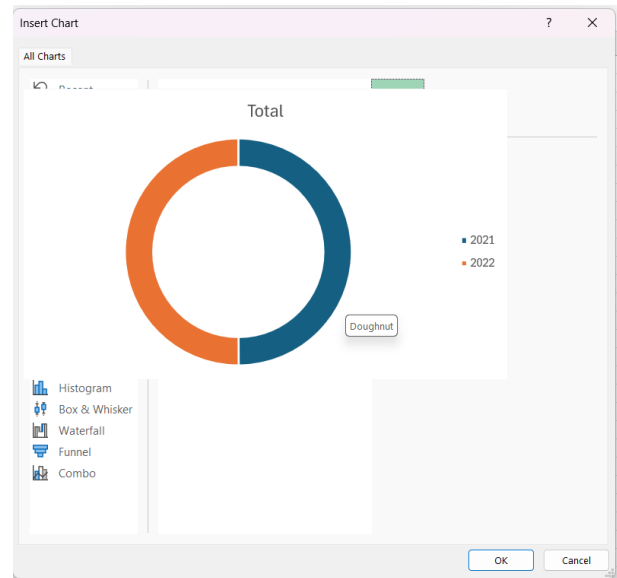
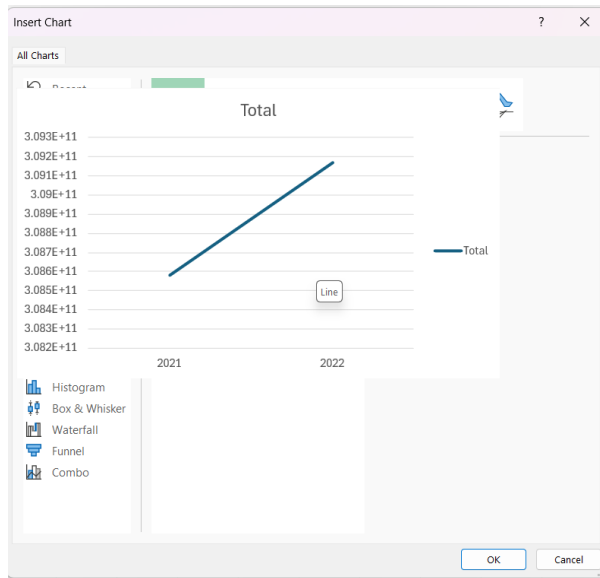
Step 3: Demonstration of OLAP operations

An Excel workbook was connected to the SSAS cube using the Data Connection Wizard. Pivot Tables were created to analyze the data using OLAP operations.

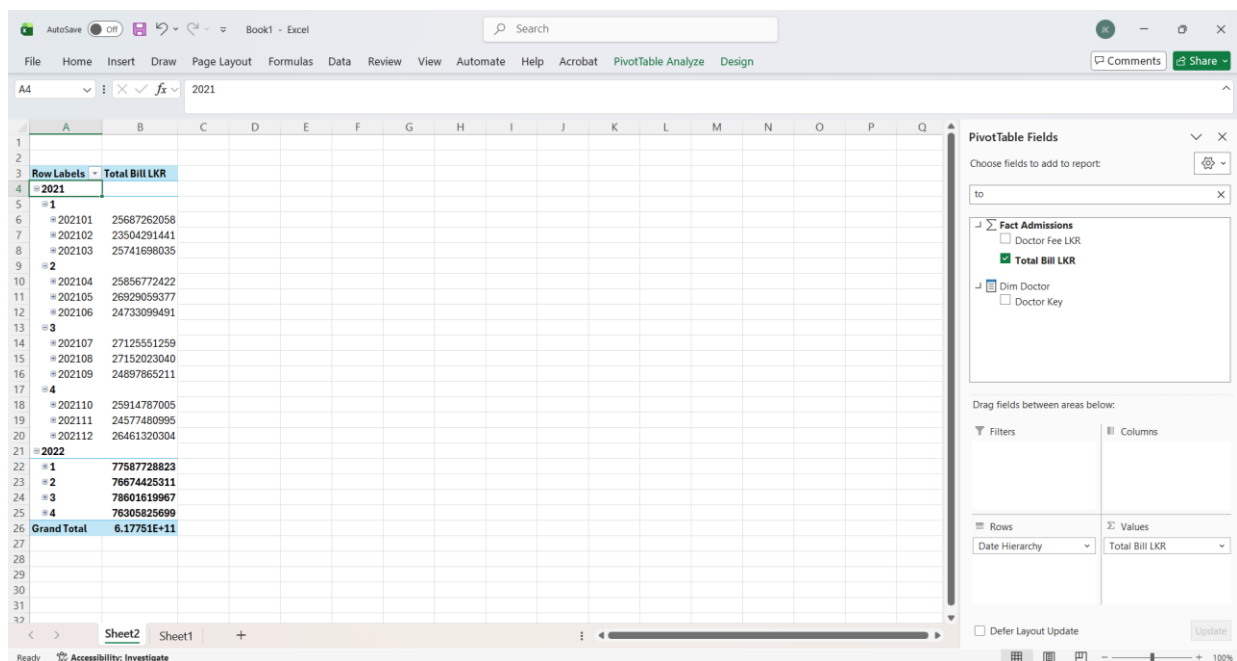
Figure: Roll-up operation

Roll-up operation was performed by collapsing the hierarchy to the Year level. This summarized the Total Bill LKR for each year, providing a high-level overview.

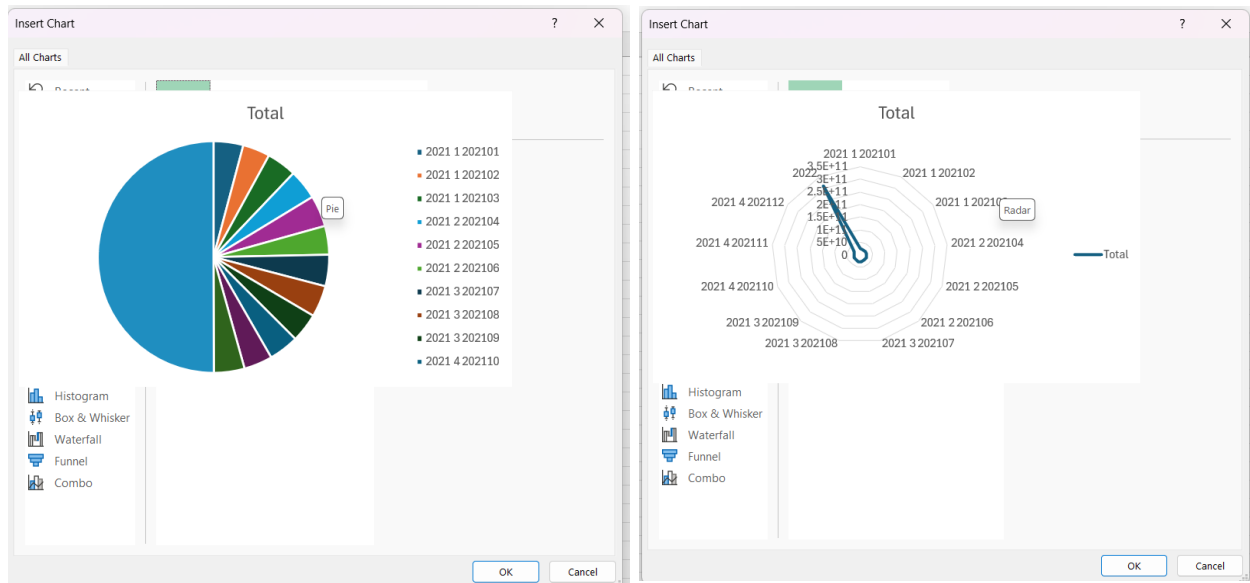




DRILL-DOWN OPERATION

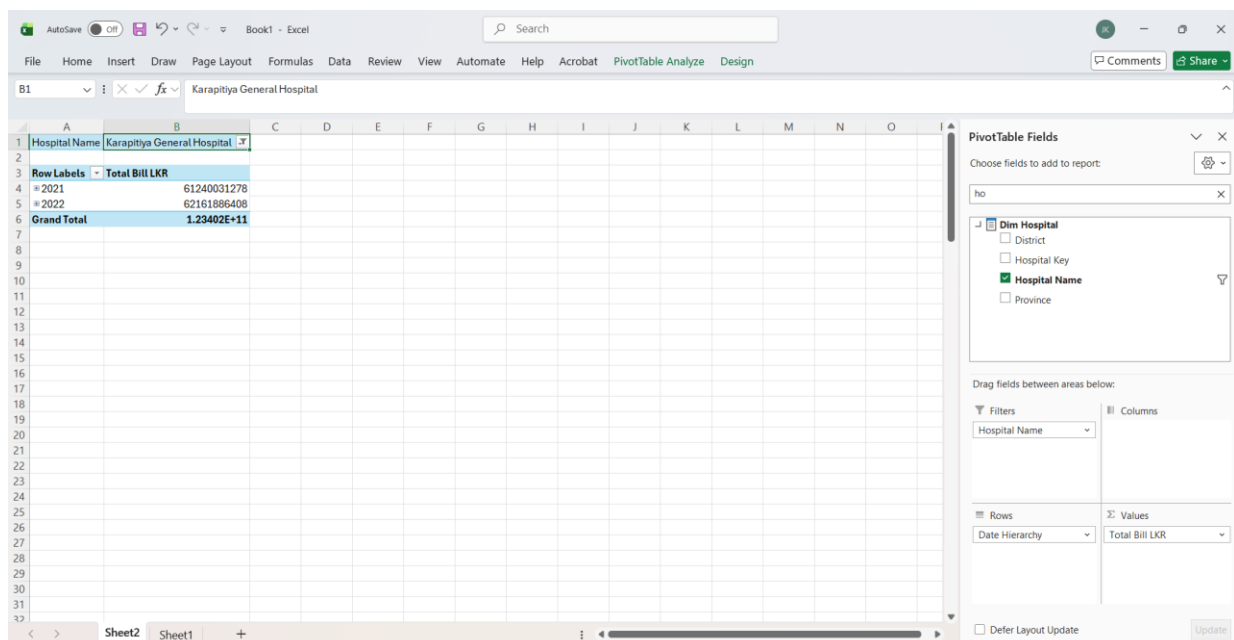


The Date dimension contains one record per month, therefore when drilling down to the Date level, only one date (the first day of each month) is displayed. This is because the dataset is aggregated at the monthly level rather than daily level.

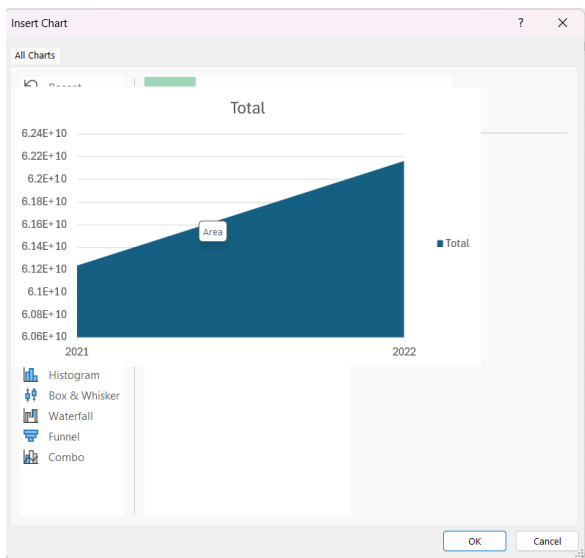
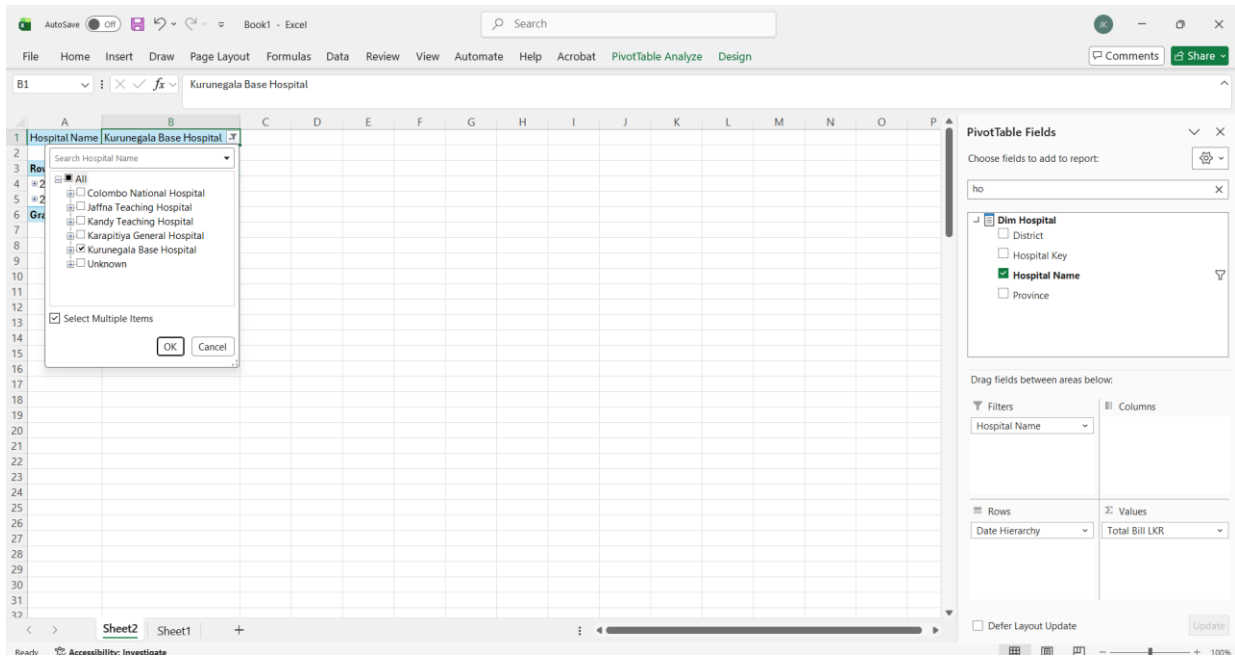


SLICE

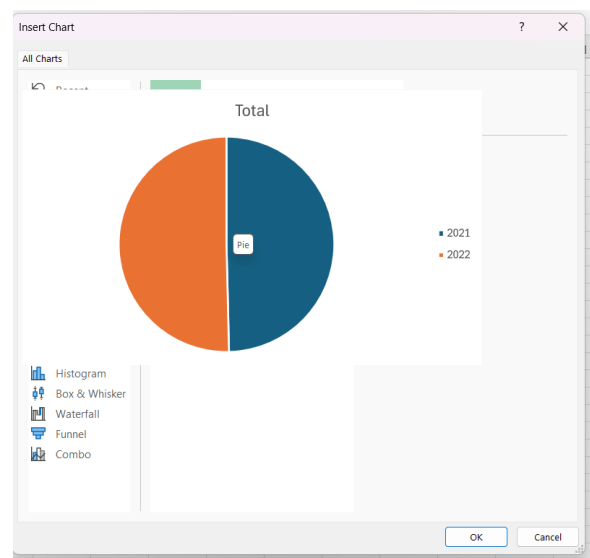
Slice operation was performed by selecting a single value from a dimension (e.g., Hospital Name) using the filter option in the Pivot Table. This allowed analysis of Total Bill LKR for a specific hospital.



Slice operation was performed by applying a filter on the Hospital Name dimension in the Pivot Table. By selecting “Karapitiya General Hospital”, the Total Bill LKR was analyzed specifically for that hospital, reducing the dataset to a focused subset.



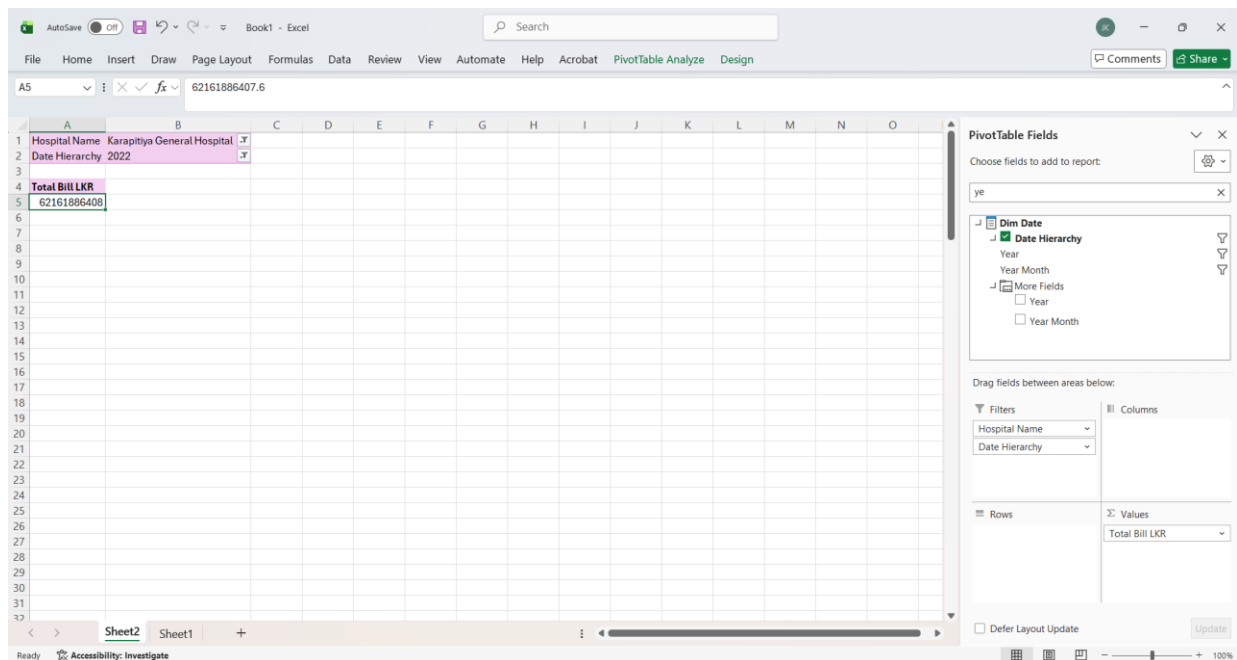
Pivot chart : Area



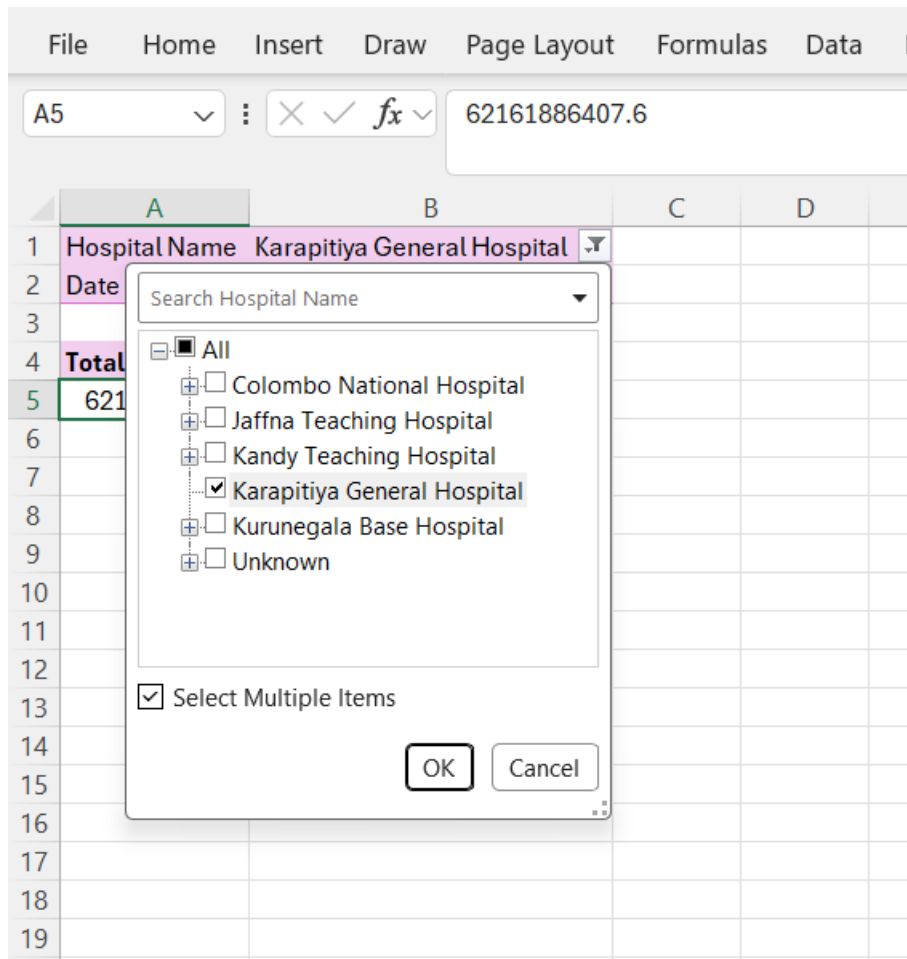
Pivot chart : Pie chart

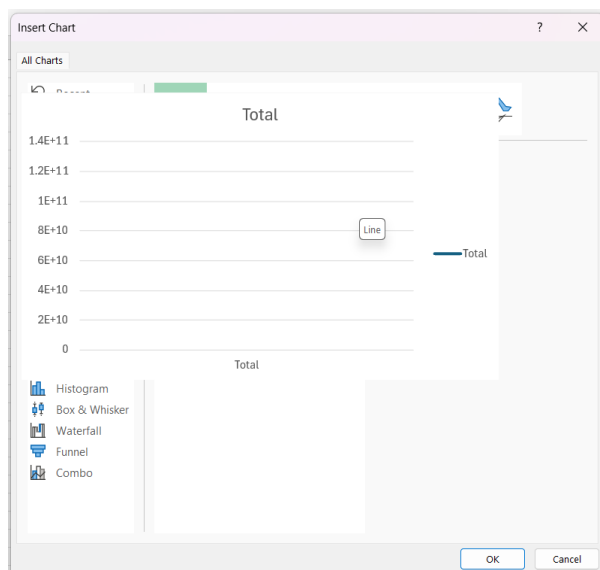
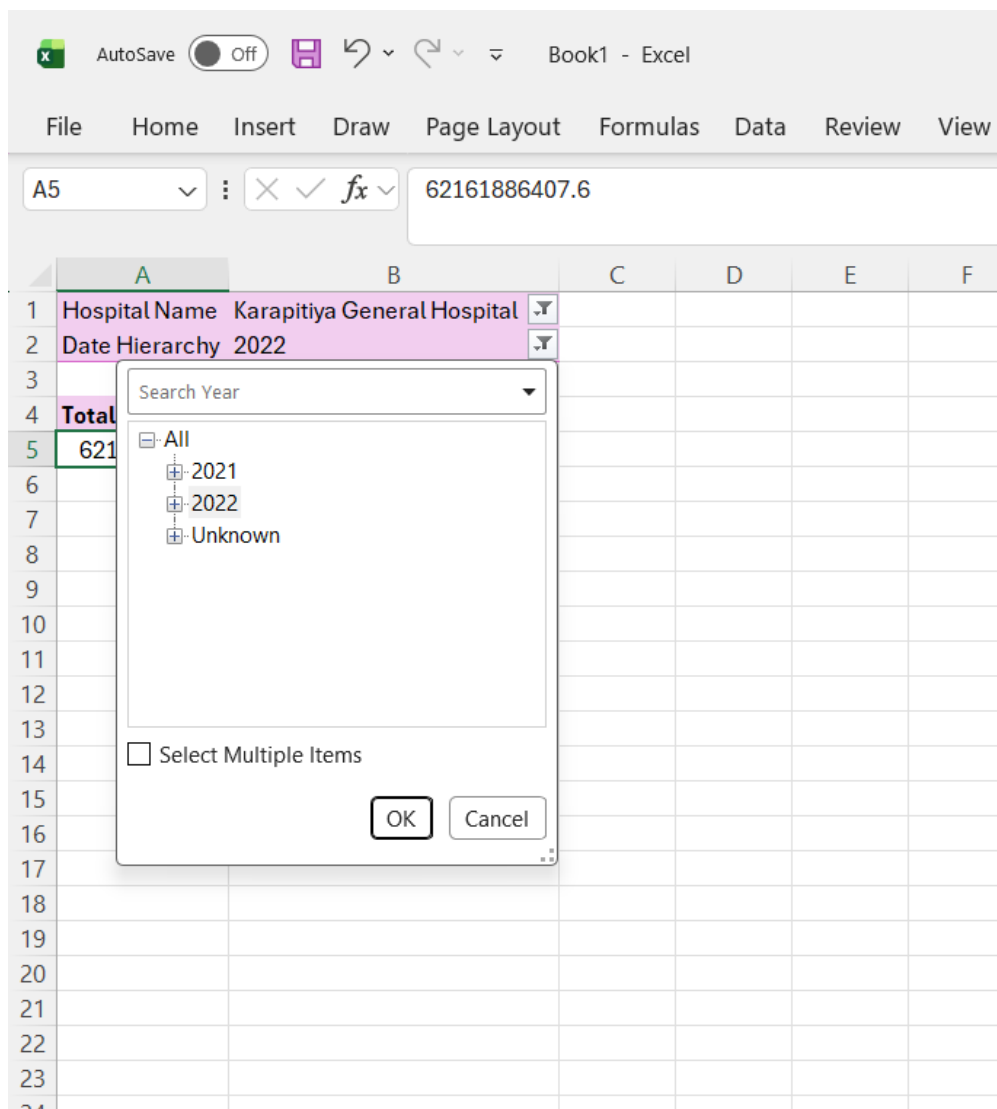
DICE (MULTIPLE FILTERS)

Dice operation was performed by applying multiple filters on different dimensions. Hospital Name and Year were selected to analyze Total Bill LKR for a specific hospital within a specific time period.

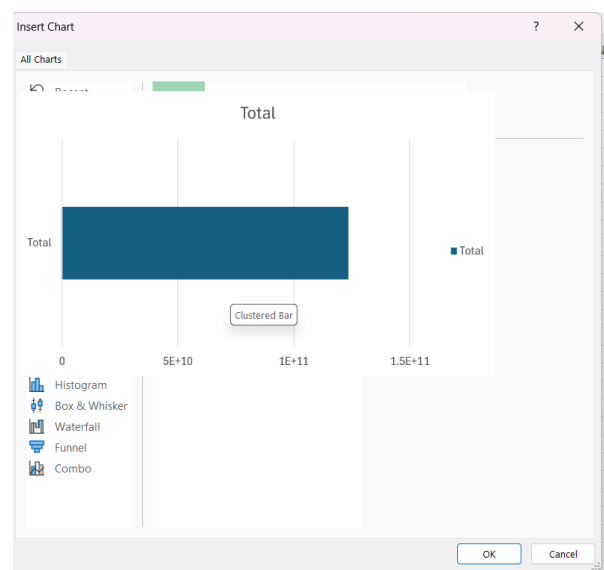


Dice operation was performed by applying multiple filters on different dimensions in the Pivot Table. Hospital Name and Year were selected to analyze Total Bill LKR for Karapitiya General Hospital in the year 2022. This allowed a more specific and detailed subset of data to be analyzed.





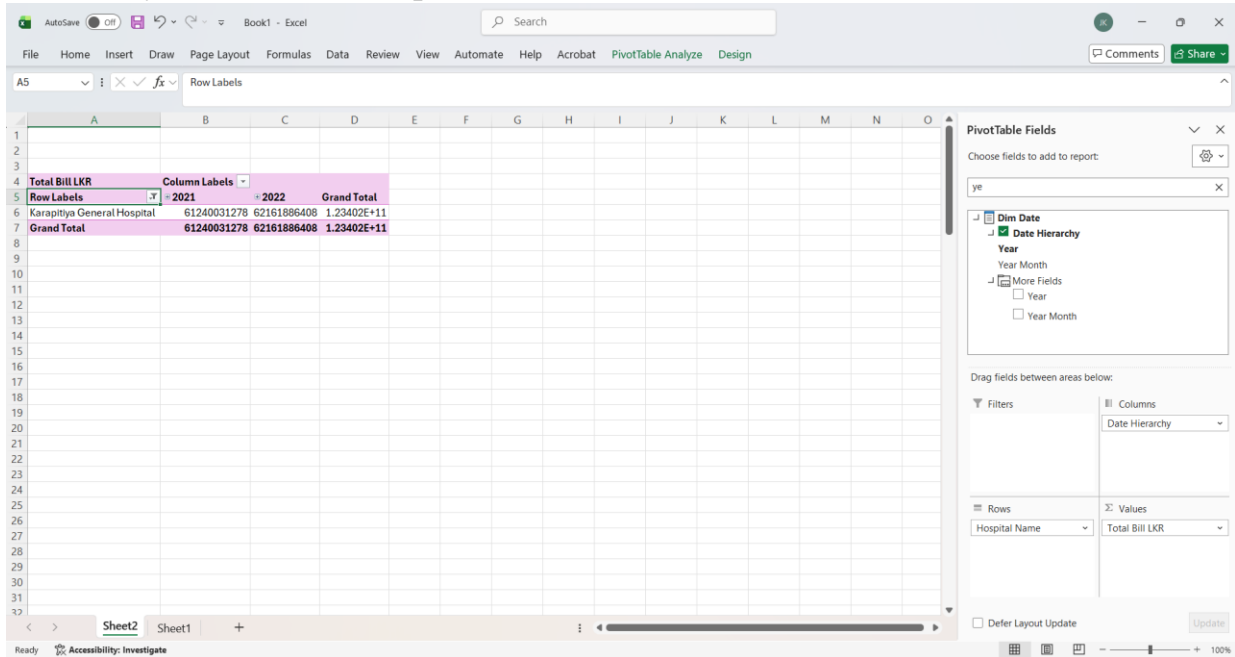
Pivot chart : Line chart



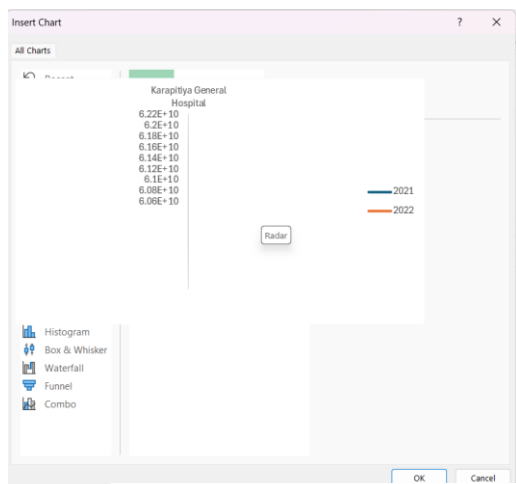
Pivot chart : Bar chart

PIVOT

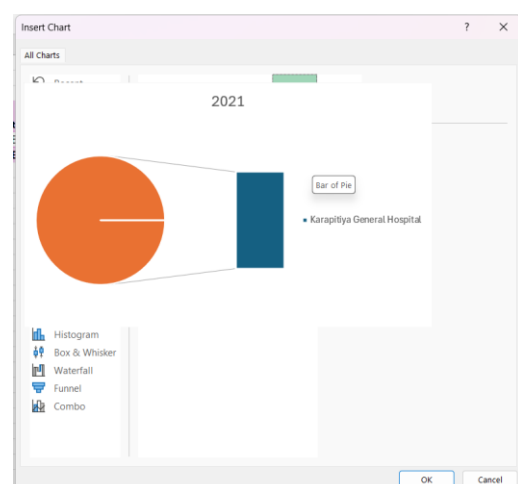
Pivot operation was performed by changing the orientation of the data in the Pivot Table. The Date Hierarchy was moved to columns and Hospital Name to rows, allowing comparison of Total Bill LKR across different years for each hospital.



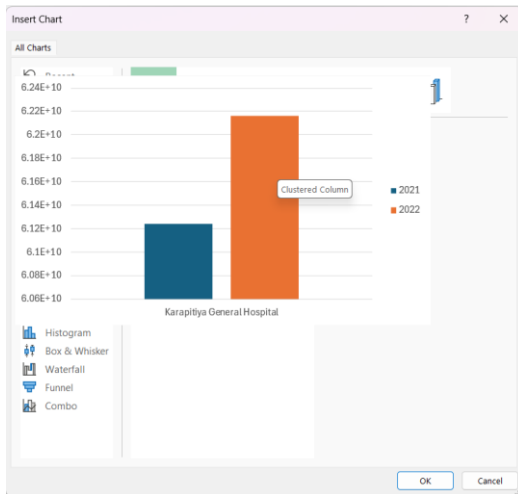
Using OLAP operations such as Roll-up, Drill-down, Slice, Dice, and Pivot, the hospital data was analyzed from multiple perspectives. These operations enabled both high-level summaries and detailed insights, demonstrating the power of multidimensional analysis using SSAS and Excel.



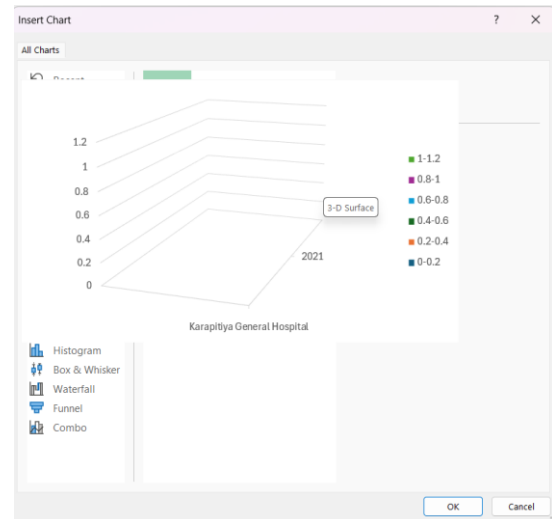
Pivot chart : Radar



Pivot chart : Bar of pie



Pivot chart : Clustered column



Pivot chart : 3- D Surface

Step 4: PowerBI Reports

Navigator

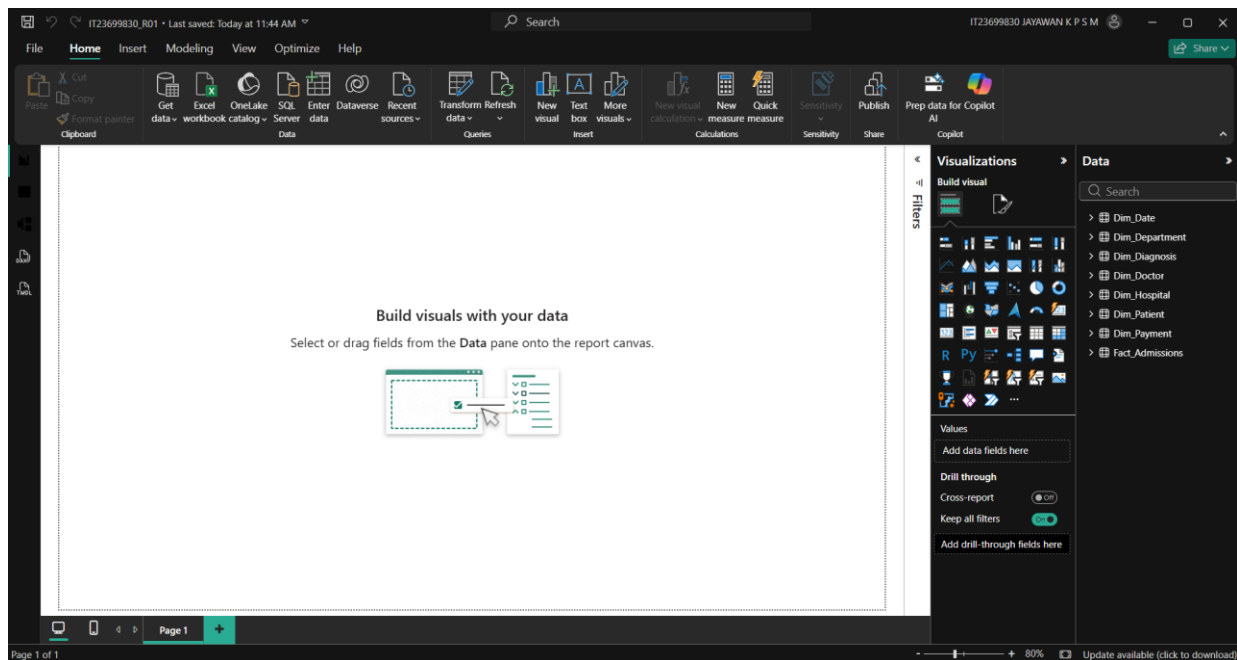
Display Options

- localhost\MSSQLSERVER2: HospitalDW [11]
 - ☒ Dim_Date
 - ☒ Dim_Department
 - ☒ Dim_Diagnosis
 - ☒ Dim_Doctor
 - ☒ Dim_Hospital
 - ☒ Dim_Patient
 - ☒ Dim_Payment
 - ☐ DimDate
 - ☒ Fact_Admissions
 - ☐ sysdiagrams
 - ☐ fn_diagramobjects

Dim_Date

Time_Key	Date	Month	Year	Quarter	Weekday	YearM
1	1/1/2021	1	2021	1	Friday	
2	1/2/2021	1	2021	1	Saturday	
3	1/3/2021	1	2021	1	Sunday	
4	1/4/2021	1	2021	1	Monday	
5	1/5/2021	1	2021	1	Tuesday	
6	1/6/2021	1	2021	1	Wednesday	
7	1/7/2021	1	2021	1	Thursday	
8	1/8/2021	1	2021	1	Friday	
9	1/9/2021	1	2021	1	Saturday	
10	1/10/2021	1	2021	1	Sunday	
11	1/11/2021	1	2021	1	Monday	
12	1/12/2021	1	2021	1	Tuesday	
13	1/13/2021	1	2021	1	Wednesday	
14	1/14/2021	1	2021	1	Thursday	
15	1/15/2021	1	2021	1	Friday	
16	1/16/2021	1	2021	1	Saturday	
17	1/17/2021	1	2021	1	Sunday	
18	1/18/2021	1	2021	1	Monday	
19	1/19/2021	1	2021	1	Tuesday	
20	1/20/2021	1	2021	1	Wednesday	
21	1/21/2021	1	2021	1	Thursday	
22	1/22/2021	1	2021	1	Friday	
23	1/23/2021	1	2021	1	Saturday	

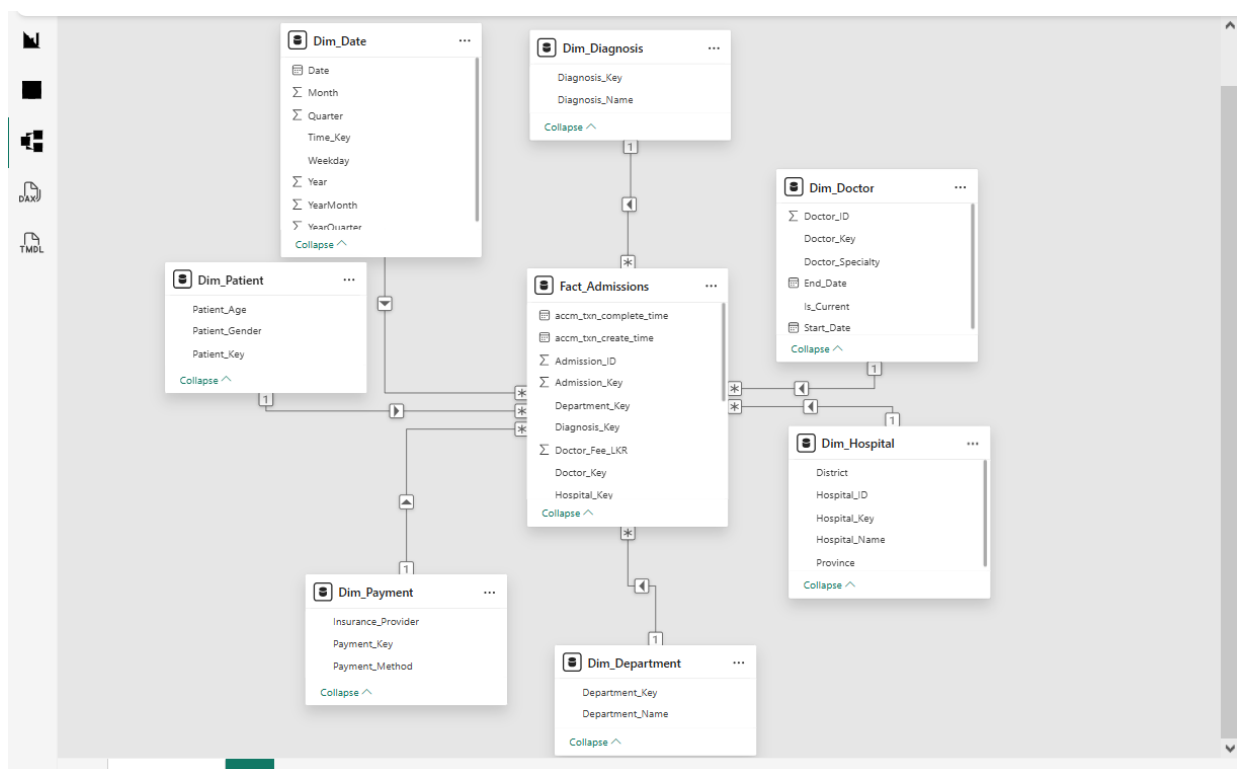
Select Related Tables Load Transform Data Cancel



THE MODEL

DATA PREPARATION

Before loading data into Power BI, the dataset was prepared by ensuring correct data types, removing inconsistencies, and validating relationships between tables. Date fields were formatted properly to support time-based analysis.



In Power BI, a data model was created by loading the data warehouse tables, including one fact table (Fact_Admissions) and multiple dimension tables such as Dim_Date, Dim_Hospital, Dim_Doctor, Dim_Patient, Dim_Department, Dim_Diagnosis, and Dim_Payment.

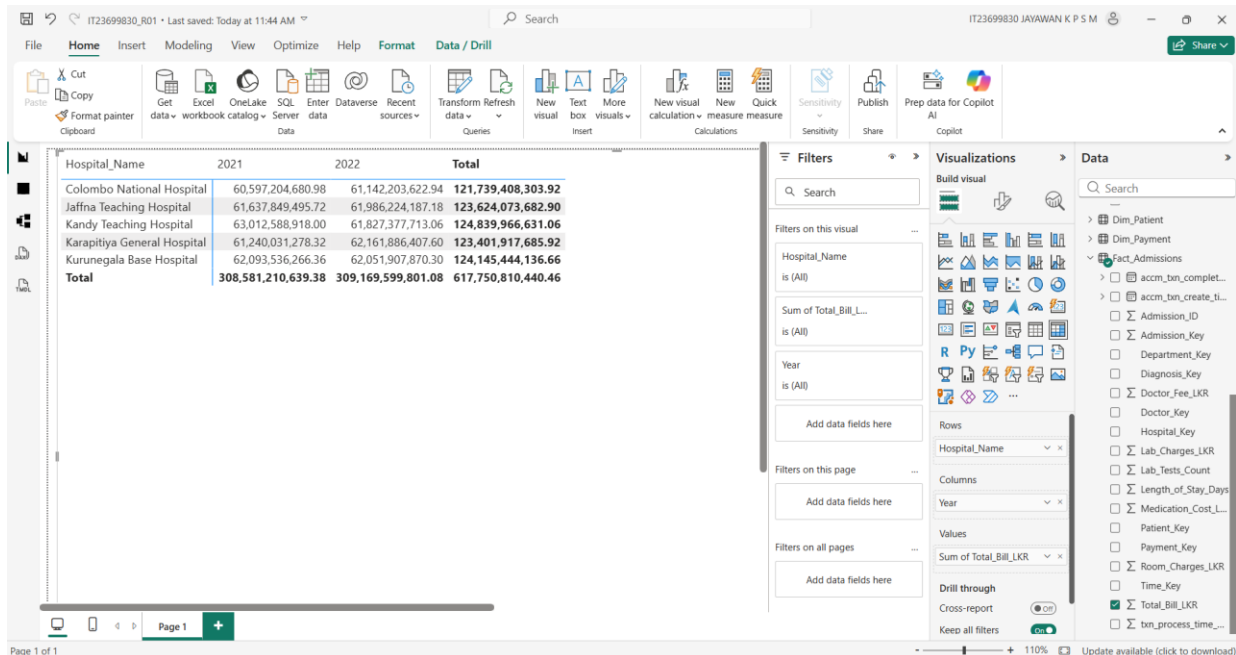
Relationships were established between the fact table and each dimension table using primary and foreign keys. The Fact_Admissions table contains foreign keys that reference the corresponding primary keys in each dimension table (e.g., Time_Key, Hospital_Key, Doctor_Key).

A star schema design was implemented, where the fact table is placed at the center and all dimension tables are connected to it. All relationships were defined as many-to-one (*:1), with the fact table on the many side and dimension tables on the one side.

This data model enables efficient filtering, slicing, and aggregation of data, allowing interactive analysis using visuals such as matrix tables, charts, slicers, and drill-down reports in Power BI.

report with a Matrix visual

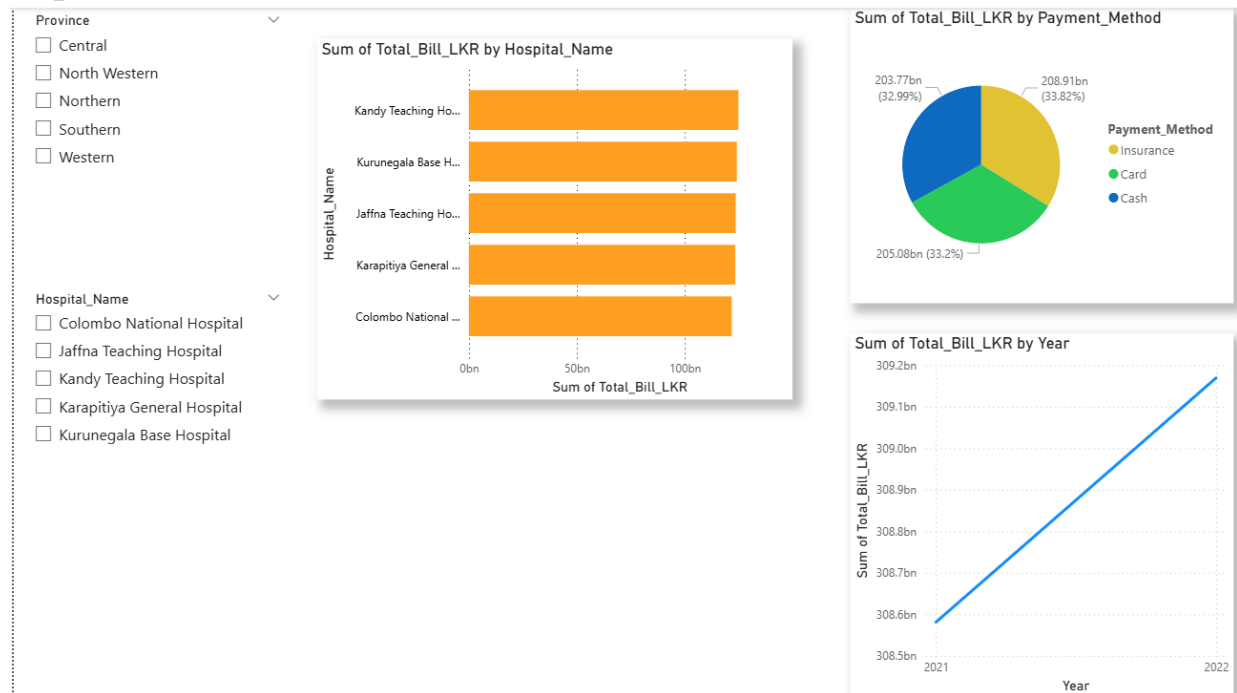
Report 1



Report 1 was created using a Matrix visual in Power BI to display detailed tabular data with row and column groupings. The Hospital Name from the Dim_Hospital table was used as rows, while the Year from the Dim_Date table was used as columns. The measure Total_Bill_LKR from the Fact_Admissions table was added as values and aggregated using SUM.

This visualization enables comparison of hospital revenue across different years and provides a structured and interactive view of the data. The Matrix visual supports analytical operations such as filtering and sorting, making it suitable for detailed data exploration.

Report 2

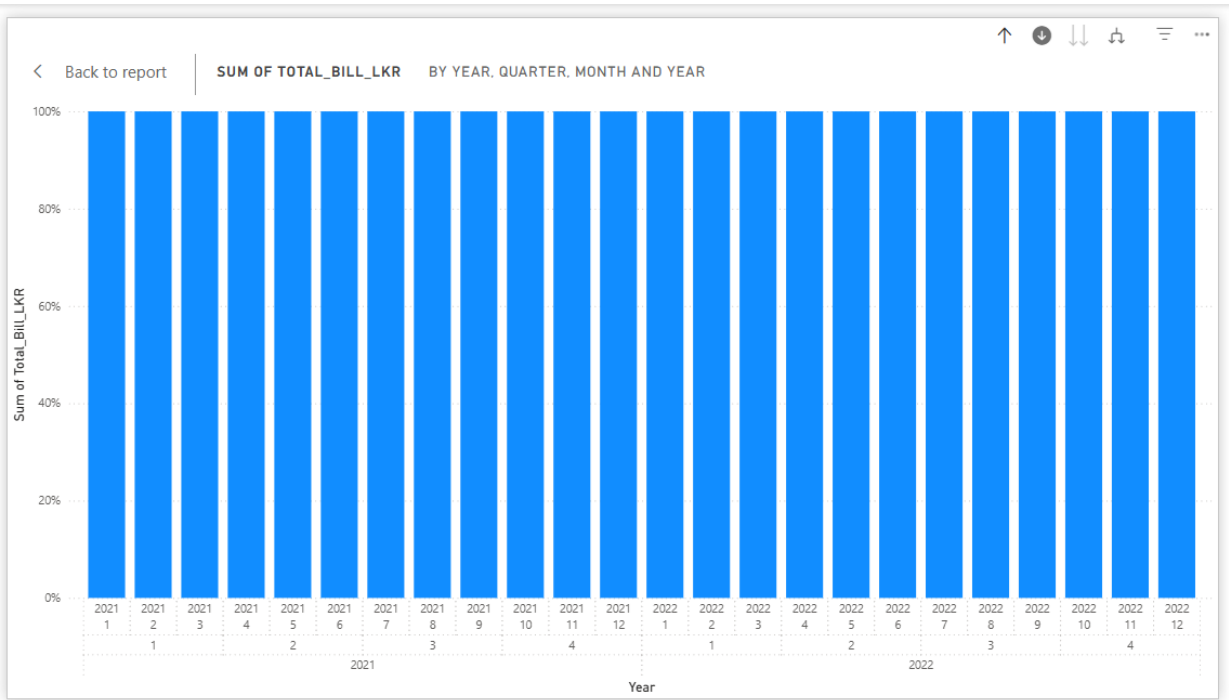
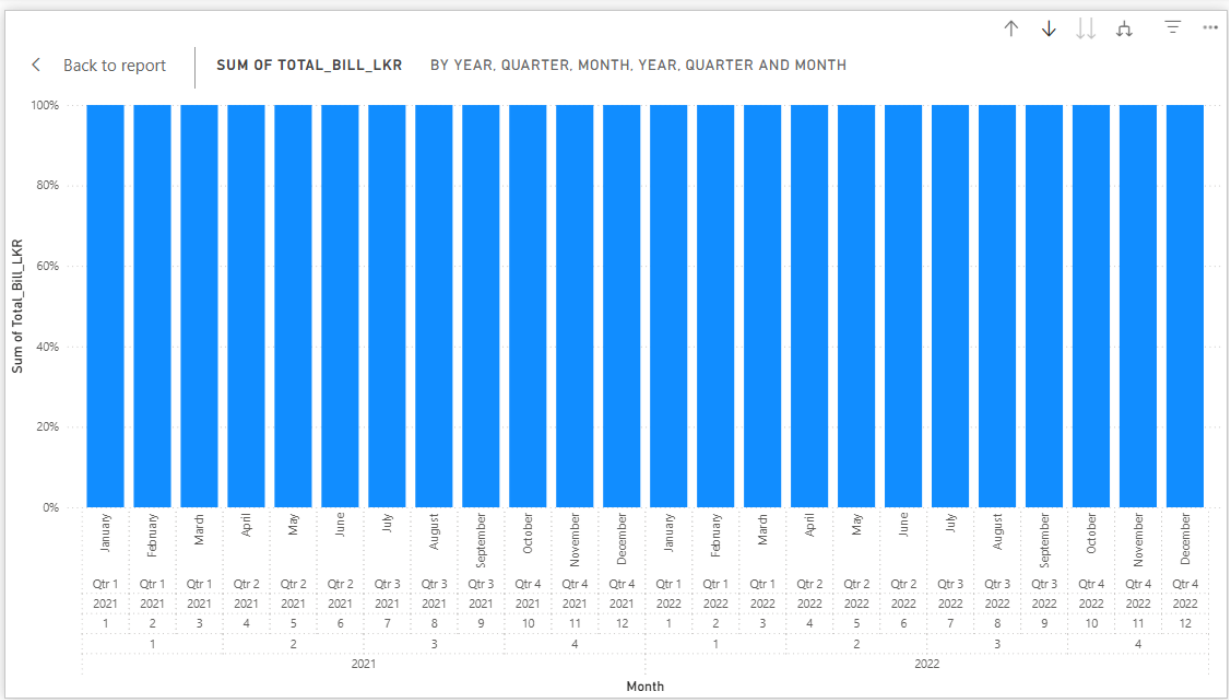


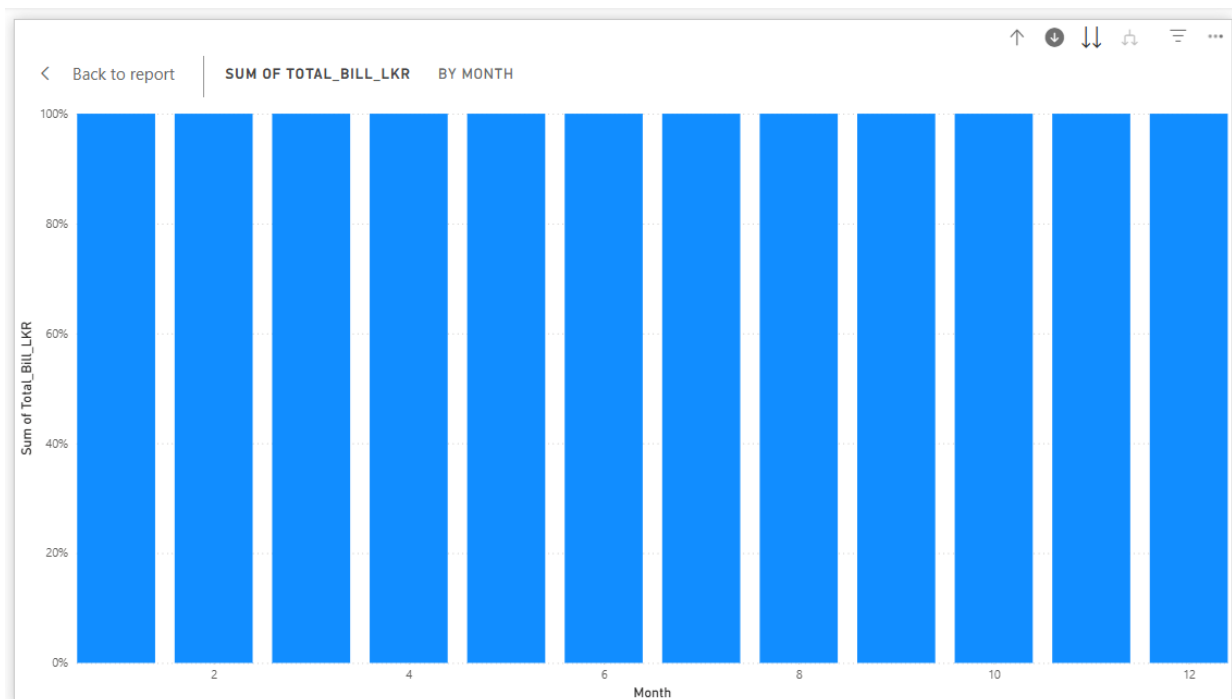
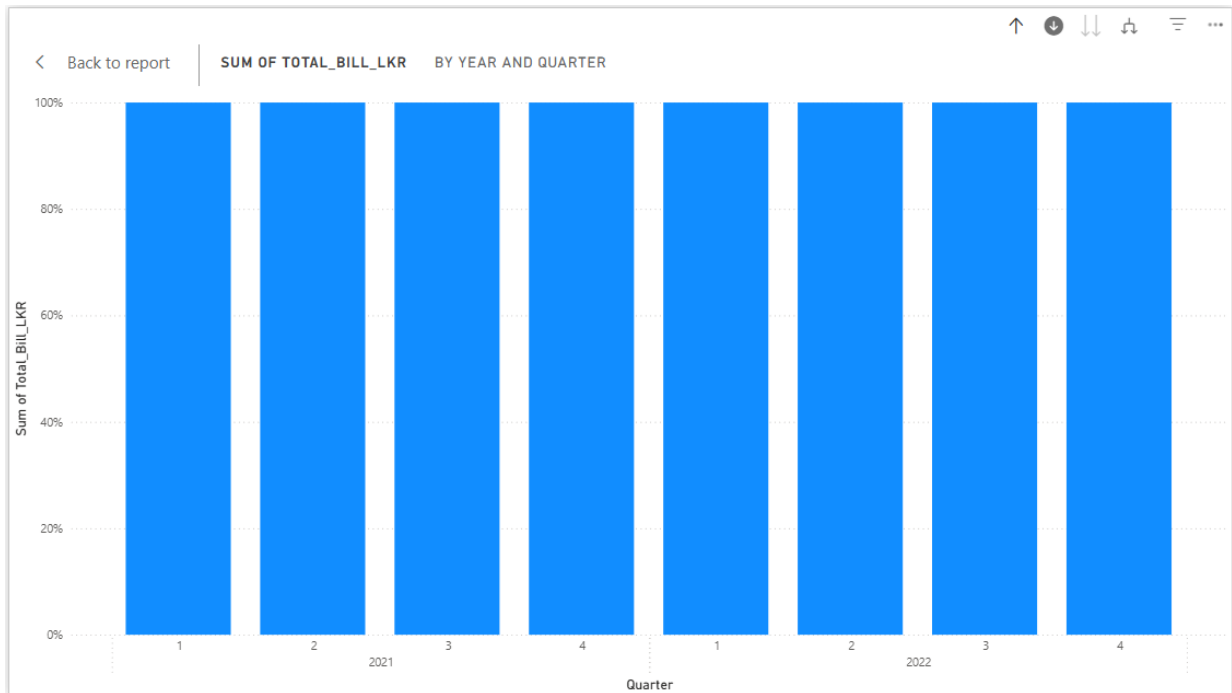
Report 2 was developed using multiple slicers and interactive visualizations in Power BI. A slicer for Province and another slicer for Hospital Name were created from the Dim_Hospital table. These slicers were configured as cascading filters, where selecting a province dynamically filters the available hospital options.

Multiple visuals were used to present insights from the dataset. A bar chart was used to compare total billing across hospitals, while a pie chart was used to visualize the distribution of billing by payment method. A line chart was created to analyze trends in total billing over time based on the Year attribute.

All visuals are interactive and respond to slicer selections, allowing users to explore the data dynamically and gain meaningful insights.

Report 3





Report 3 was created to demonstrate drill-down functionality using hierarchical data in Power BI. A column chart was used to visualize total billing amounts based on a time hierarchy.

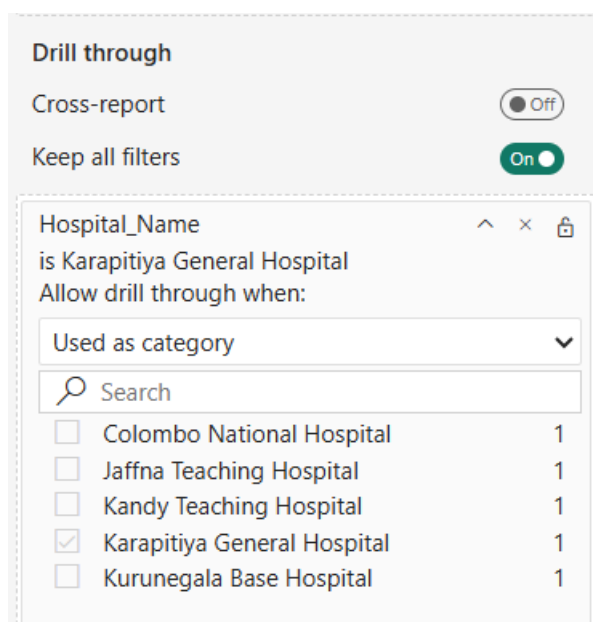
A hierarchy was created using the Dim_Date table, including Year, Quarter, Month, and Date. These fields were added to the axis of the chart, while Total_Bill_LKR from the Fact_Admissions table was added to the values. Drill-down functionality was enabled, allowing users to interactively navigate through different levels of the hierarchy. Users can start by viewing data at the yearly level and drill down into quarters, months, and individual dates.

This approach enables detailed analysis and improves the ability to explore trends and patterns at multiple levels of granularity.

Report 4

Report 4 was developed using Power BI to demonstrate advanced analytical features including matrix visualization, cascading slicers, drill-down, and drill-through functionality. The report enables users to interactively explore hospital data and gain insights into billing, departments, and payment methods.

Drill-through setup



Hospital_Name	2021	2022	Total
Colombo National Hospital	60,597,204,680.98	61,142,203,622.94	121,739,408,303.92
Jaffna Teaching Hospital	61,637,849,495.72	61,986,224,187.18	123,624,073,682.90
Kandy Teaching Hospital	63,012,588,918.00	61,827,377,713.06	124,839,966,631.06
Karapitiya General Hospital	61,248,231,278.32	62,161,886,407.60	123,401,917,685.92
Kurunegala General Hospital	61,366,266.36	62,051,907,870.30	124,145,444,136.66
Total	309,0639.38	309,169,599,801.08	617,750,810,440.46

Copy
Expand
Collapse
Show as a table
Include
Exclude
Drill through
Group
Clear selections
Summarize
Unfreeze row headers
New visual calculation
Set up a verified answer

Report 4 – Drill Through

Drill through

Drill-through result page

Hospital_Name	Sum of Year	Sum of Total_Bill_LKR
Karapitiya General Hospital	1475695	123,401,917,685.92
Total	1475695	123,401,917,685.92

Sum of Total_Bill_LKR by Department_Name

Filters
Visualizations

Filters on this page
Add data fields here
Filters on all pages
Add data fields here

Values
Add data fields here
Drill through
Cross-report
Keep all filters
Hospital_Name is Karapitiya General Hospital

Report 4 – Drill Through

Report 4 was developed to demonstrate the drill-through functionality in Power BI, which allows users to navigate from a summary report to a detailed report page for deeper analysis.

A new report page was created and configured as a drill-through page. The attribute Hospital_Name from the Dim_Hospital table was added to the drill-through filter section. This enables users to right-click on a hospital in other report pages and navigate to this detailed view.

The drill-through page was designed to display detailed information related to the selected hospital. A table visual was used to present data such as Hospital_Name, Year, and Total_Bill_LKR. In addition, a bar chart was created to show department-wise total billing, allowing users to analyze how different departments contribute to the overall hospital revenue.

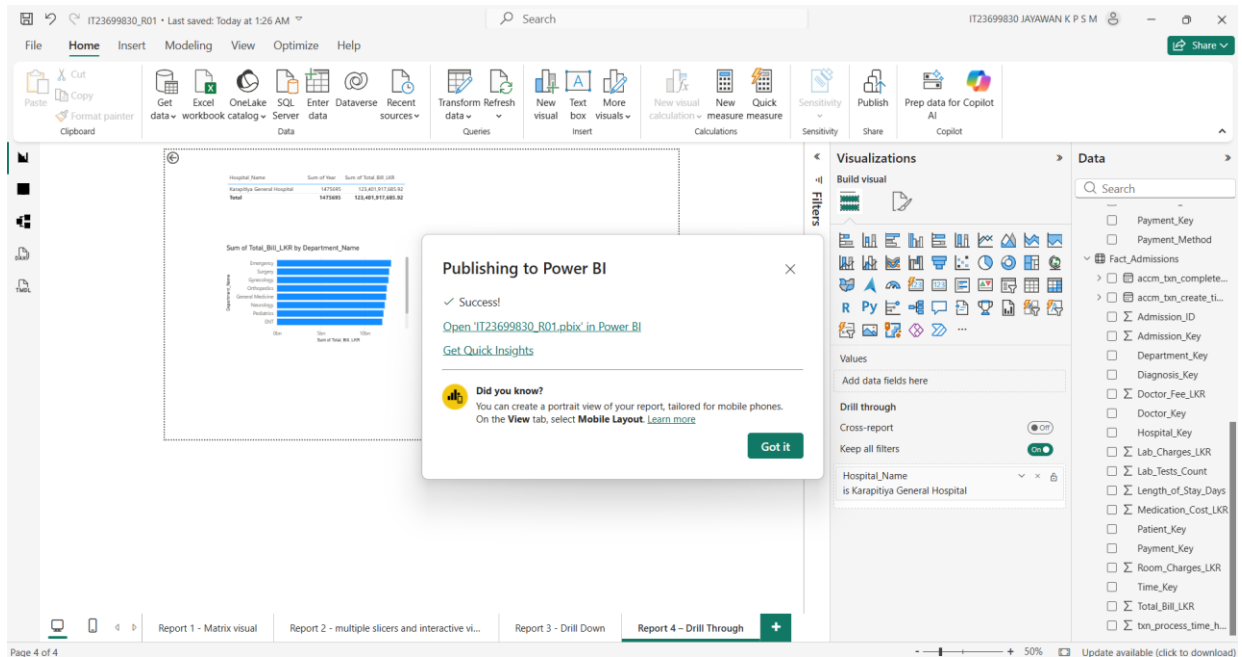
To improve usability, a Back button was added and configured with the “Back” action. This allows users to return to the original report page after performing drill-through navigation. In Power BI Desktop, Ctrl + Click is required to activate drill-through and back navigation, while in Power BI Service, normal clicks are sufficient.

This drill-through feature enhances user interaction by enabling detailed exploration of specific data points directly from summary visuals, making the report more dynamic and insightful.

DAX USAGE

No complex DAX measures were created in this project, as built-in aggregation functions such as SUM were sufficient for analysis. However, the data model supports future enhancements using DAX for advanced calculations.

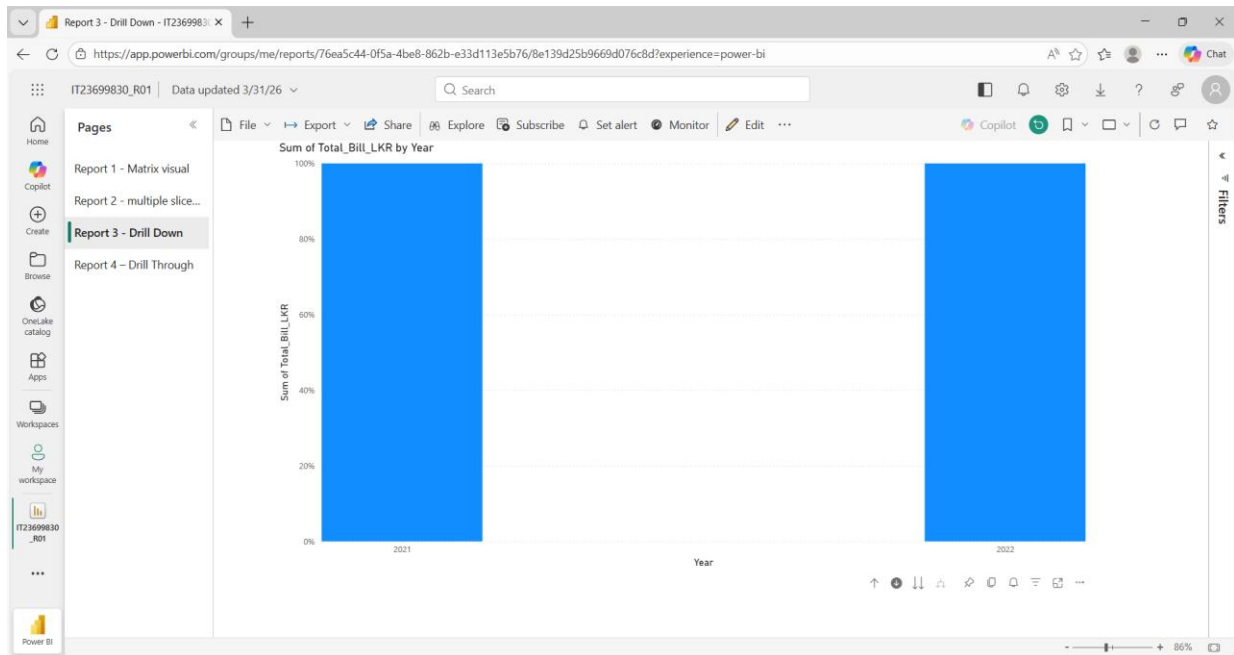
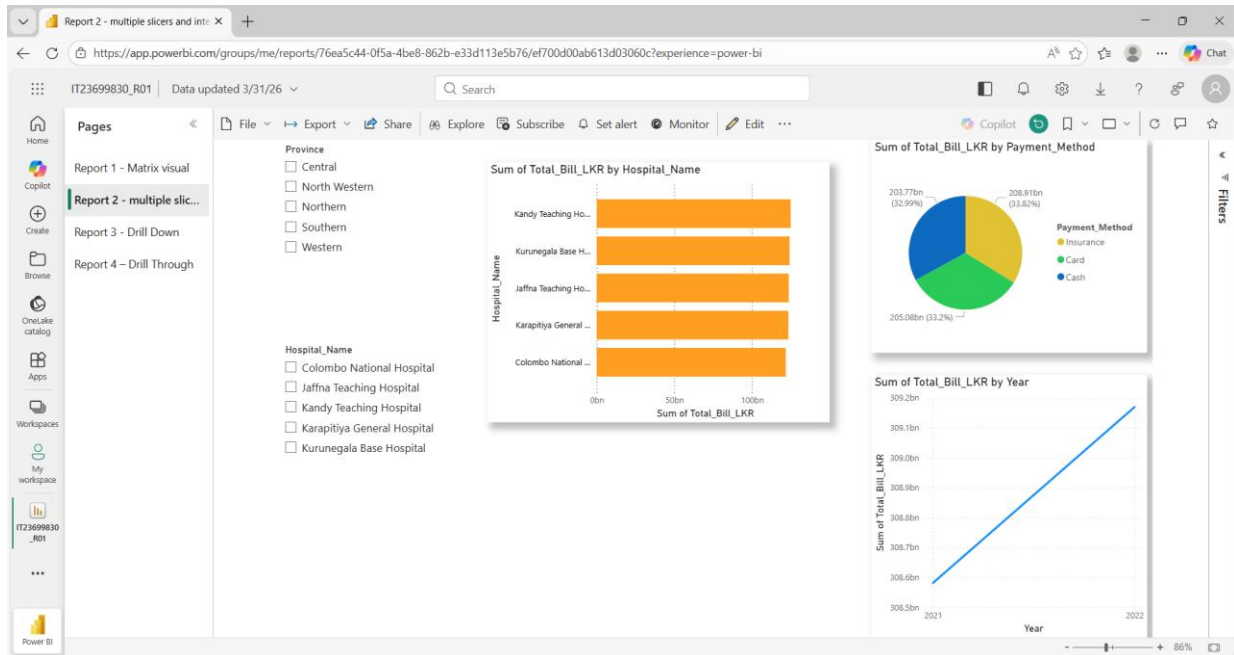
Publishing to PowerBI

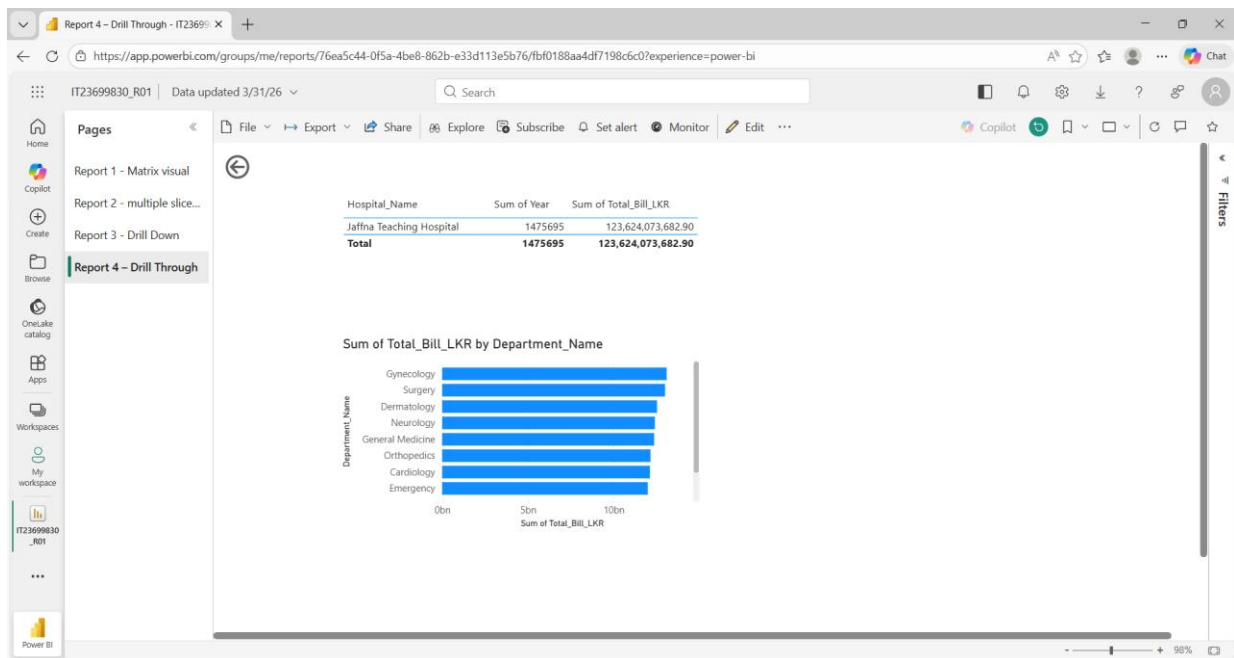


Link - <https://app.powerbi.com/groups/me/reports/76ea5c44-0f5a-4be8-862b-e33d113e5b76/fb0188aa4df7198c6c0?experience=power-bi>

The screenshot shows the Power BI service web interface. The report is titled 'Report 1 - Matrix visual' and displays a matrix with columns for Hospital_Name, 2021, 2022, and Total. The data is the same as shown in the Power BI Desktop screenshot.

Hospital_Name	2021	2022	Total
Jayanti Teaching Hospital	60,597,204,680.98	61,142,203,622.94	121,739,408,303.92
Kandy Teaching Hospital	61,637,849,495.72	61,986,224,187.16	123,624,073,682.90
Karapitiya General Hospital	63,012,588,918.00	61,827,377,713.06	124,839,966,631.06
Kurumegala Base Hospital	61,240,031,278.32	62,161,886,407.60	123,401,917,685.92
Total	62,093,536,266.36	62,051,907,870.30	124,145,444,136.66





The Power BI reports successfully demonstrate interactive data analysis using matrix visuals, slicers, drill-down, and drill-through operations. These features enhance data exploration and support better decision-making.